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Reg no : 23MIS0178

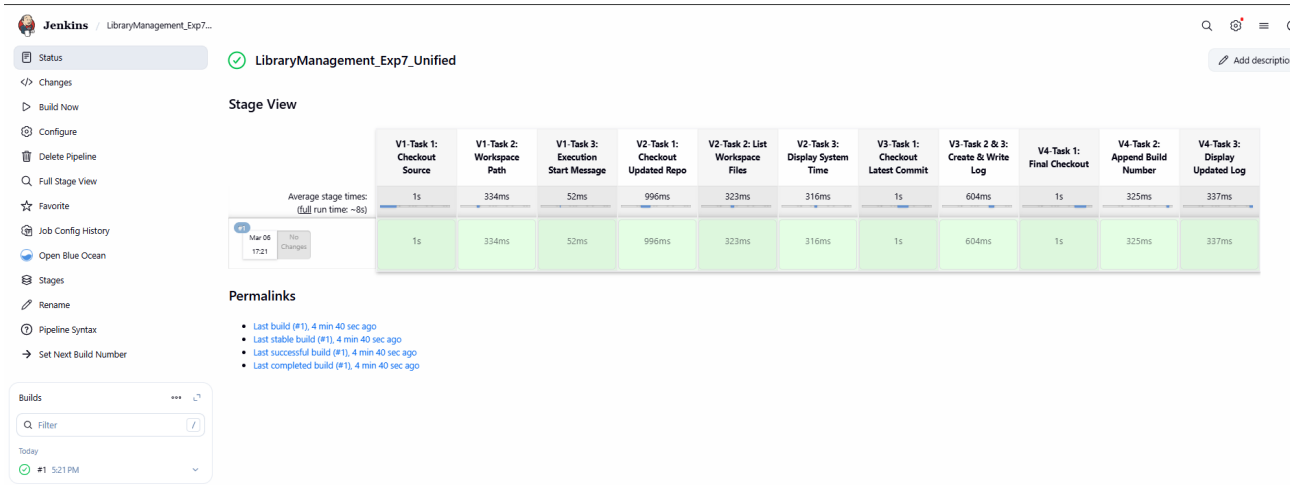
Experiment :7,8

Experiment 7: Workspace Information and Logging Objective: To understand Jenkins workspace information and logging using simple pipeline commands.

Script:

```
pipeline {
    agent any

    stages {
        // --- VERSION 1: INITIAL SETUP & WORKSPACE ---
        stage('V1-Task 1: Checkout Source') {
            steps {
                git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
            }
        }
        stage('V1-Task 2: Workspace Path') {
            steps {
                bat 'echo The current Jenkins workspace is: %WORKSPACE%'
            }
        }
        stage('V1-Task 3: Execution Start Message') {
            steps {
                echo "Pipeline execution has started."
            }
        }

        // --- VERSION 2: DIRECTORY & TIME ---
        stage('V2-Task 1: Checkout Updated Repo') {
            steps {
                git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
            }
        }
        stage('V2-Task 2: List Workspace Files') {
            steps {
                echo "Listing all files and folders:"
                bat 'dir'
            }
        }
        stage('V2-Task 3: Display System Time') {
            steps {
                bat 'echo Current System Time: & time /t'
            }
        }

        // --- VERSION 3: LOG CREATION ---
        stage('V3-Task 1: Checkout Latest Commit') {
```

```
        steps {
            git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
        }
    }
    stage('V3-Task 2 & 3: Create & Write Log') {
        steps {
            // ">" creates the file and writes the initial message
            bat 'echo Build initiated by Jenkins > buildlog.txt'
            echo "Displaying buildlog.txt contents:"
            bat 'type buildlog.txt'
        }
    }

    // --- VERSION 4: LOG APPENDING ---
    stage('V4-Task 1: Final Checkout') {
        steps {
            git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
        }
    }
    stage('V4-Task 2: Append Build Number') {
        steps {
            // ">>" appends the build number to the end of the file
            bat 'echo Appending Build Number: %BUILD_NUMBER% >> buildlog.txt'
        }
    }
    stage('V4-Task 3: Display Updated Log') {
        steps {
            echo "Updated contents of buildlog.txt:"
            bat 'type buildlog.txt'
        }
    }
  }
}
```



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#1 (Mar 6, 2026, 5:21:18 PM)

Started by user admin

This run spent:

- 8 ms waiting;
- 8.1 sec build duration;
- 8.1 sec total from scheduled to completion.

Revision: 67abf8278758a9ec932fab3d6393d33bf988488b
Repository: https://github.com/jaswanthhitman45/LibraryManagementSystem

- refs/remotes/origin/master

No changes.

Jenkins / LibraryManagement_Exp7... / #1 / Console Output

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#1'
- Timings
- Git Build Data
- Rebuild
- Open Blue Ocean
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Next Build

Console Output

```
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\LibraryManagement_Exp7_Unified
[Pipeline] {
[Pipeline] stage
[Pipeline] { (V1-Task 1: Checkout Source)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\LibraryManagement_Exp7_Unified # timeout=10
Fetching upstream changes from https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/jaswanthhitman45/LibraryManagementSystem +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe config remote.origin.url https://github.com/jaswanthhitman45/LibraryManagementSystem # timeout=10
 > git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision 67abf8278758a9ec932fab3d6393d33bf988488b (refs/remotes/origin/master)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe checkout -b master 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
Commit message: "Added gitignore file"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (V1-Task 2: Workspace Path)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\LibraryManagement_Exp7_Unified>echo The current Jenkins workspace is: C:\ProgramData\Jenkins\.jenkins\workspace\Library
The current Jenkins workspace is: C:\ProgramData\Jenkins\.jenkins\workspace\LibraryManagement_Exp7_Unified
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
```

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ement_Exp7... / #1 / Console Output

```
 > git.exe config remote.origin.url https://github.com/jaswanthhitman45/LibraryManagementSystem # timeout=10
Fetching upstream changes from https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/jaswanthhitman45/LibraryManagementSystem +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision 67abf8278758a9ec932fab3d6393d33bf988488b (refs/remotes/origin/master)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe branch -D master # timeout=10
 > git.exe checkout -b master 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
Commit message: "Added gitignore file"
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (V4-Task 2: Append Build Number)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\LibraryManagement_Exp7_Unified>echo Appending Build Number: 1  1>>buildlog.txt
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (V4-Task 3: Display Updated Log)
[Pipeline] echo
Updated contents of buildlog.txt:
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\LibraryManagement_Exp7_Unified>type buildlog.txt
Build initiated by Jenkins
Appending Build Number: 1
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

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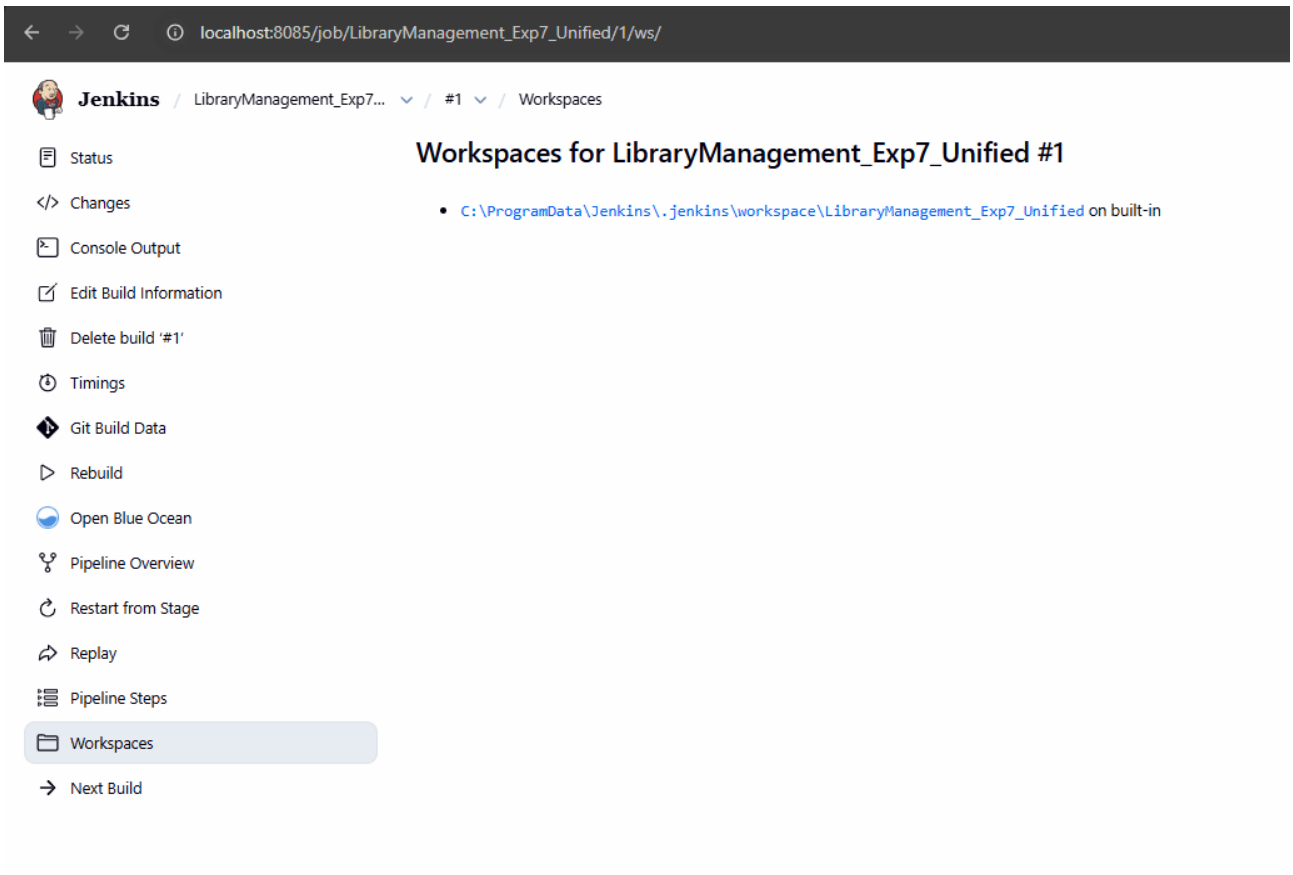


Experiment 8: Parameterized Pipeline Execution Objective: To understand how Jenkins pipelines accept parameters and use them during execution.

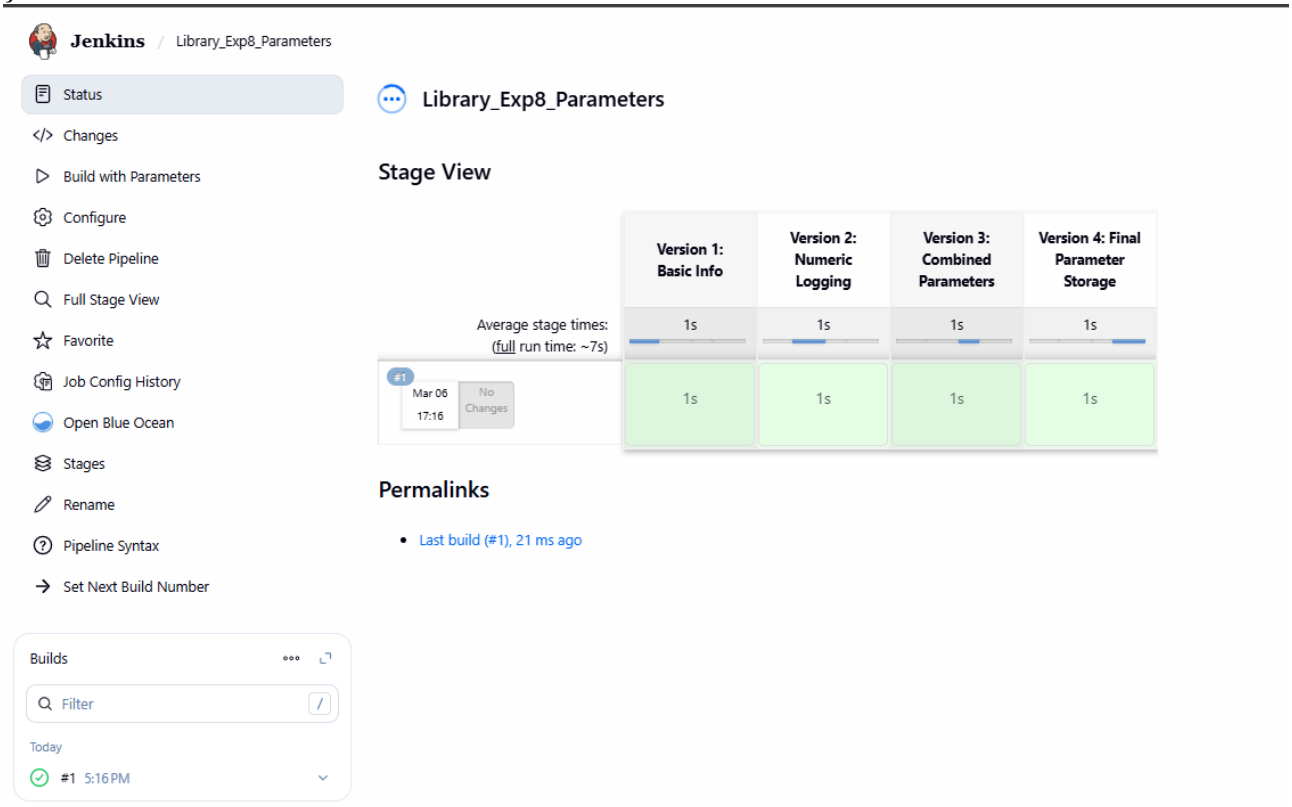
Version 1 • Task 1: Checkout the project from GitHub. • Task 2: Define a string parameter named PROJECT_NAME. • Task 3: Display the value of the parameter in the Jenkins console.

Script:

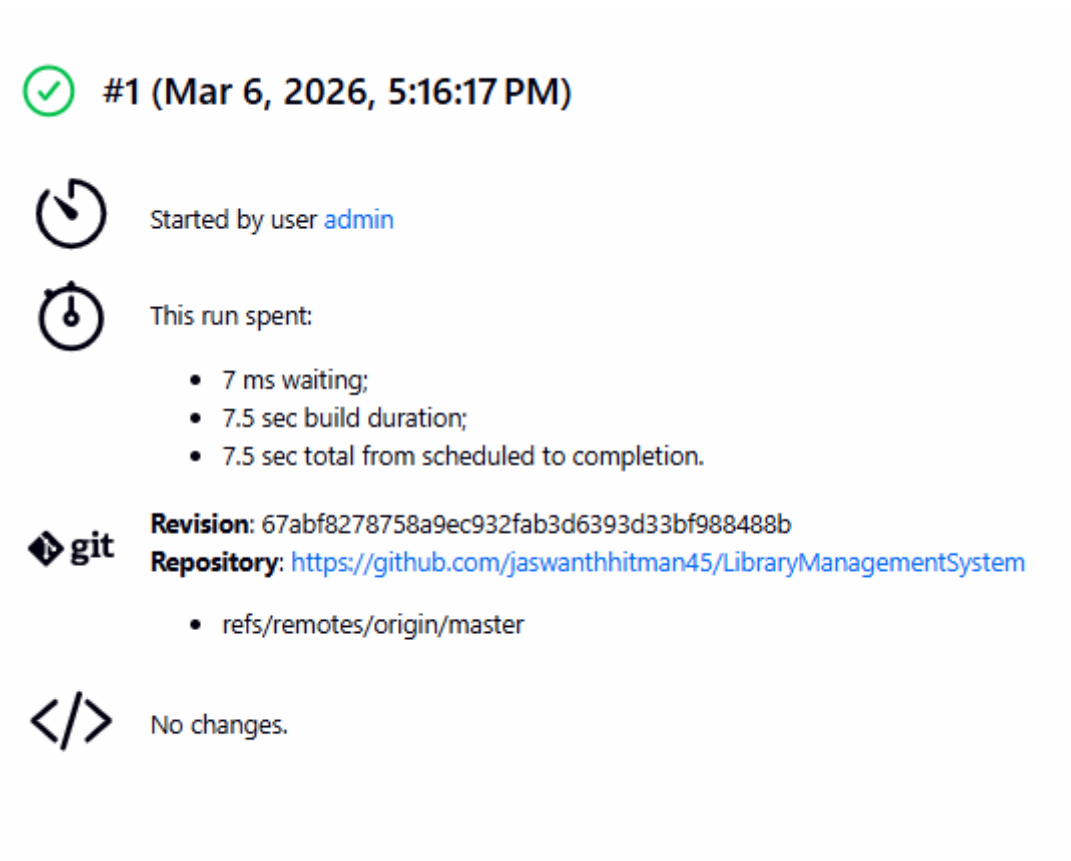
```
pipeline {
  agent any

  stages {
    // --- VERSION 1: INITIAL SETUP & WORKSPACE ---
    stage('V1-Task 1: Checkout Source') {
      steps {
        git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
      }
    }
    stage('V1-Task 2: Workspace Path') {
      steps {
        bat 'echo The current Jenkins workspace is: %WORKSPACE%'
      }
    }
    stage('V1-Task 3: Execution Start Message') {
      steps {
        echo "Pipeline execution has started."
```

```
    }
}

// --- VERSION 2: DIRECTORY & TIME ---
stage('V2-Task 1: Checkout Updated Repo') {
    steps {
        git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
    }
}
stage('V2-Task 2: List Workspace Files') {
    steps {
        echo "Listing all files and folders:"
        bat 'dir'
    }
}
stage('V2-Task 3: Display System Time') {
    steps {
        bat 'echo Current System Time: & time /t'
    }
}

// --- VERSION 3: LOG CREATION ---
stage('V3-Task 1: Checkout Latest Commit') {
    steps {
        git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
    }
}
stage('V3-Task 2 & 3: Create & Write Log') {
    steps {
        // ">" creates the file and writes the initial message
        bat 'echo Build initiated by Jenkins > buildlog.txt'
        echo "Displaying buildlog.txt contents:"
        bat 'type buildlog.txt'
    }
}

// --- VERSION 4: LOG APPENDING ---
stage('V4-Task 1: Final Checkout') {
    steps {
        git 'https://github.com/jaswanthhitman45/LibraryManagementSystem'
    }
}
stage('V4-Task 2: Append Build Number') {
    steps {
        // ">>" appends the build number to the end of the file
        bat 'echo Appending Build Number: %BUILD_NUMBER% >> buildlog.txt'
    }
}
stage('V4-Task 3: Display Updated Log') {
    steps {
        echo "Updated contents of buildlog.txt:"
        bat 'type buildlog.txt'
```

```
        }
    }
  }
}
```


Jenkins / Library_Exp8_Parameters
Status
Changes
Build with Parameters
Configure
Delete Pipeline
Full Stage View
Favorite
Job Config History
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Set Next Build Number
Builds
Filter
Today
#1 5:16 PM
Library_Exp8_Parameters
Stage View
Version 1: Basic Info
Version 2: Numeric Logging
Version 3: Combined Parameters
Version 4: Final Parameter Storage
Average stage times:
(full run time: ~7s)
1s
#1
Mar 06
17:16
No Changes
Permalinks
Last build (#1), 21 ms ago

.


#1 (Mar 6, 2026, 5:16:17 PM)
Started by user admin
This run spent:
7 ms waiting;
7.5 sec build duration;
7.5 sec total from scheduled to completion.
git
Revision: 67abf8278758a9ec932fab3d6393d33bf988488b
Repository: https://github.com/jaswanthhitman45/LibraryManagementSystem
refs/remotes/origin/master
No changes.

.

Console Output

```
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Version 1: Basic Info)
[Pipeline] echo
Task 1: Checking out source code...
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters # timeout=10
Fetching upstream changes from https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/jaswanthhitman45/LibraryManagementSystem +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe config remote.origin.url https://github.com/jaswanthhitman45/LibraryManagementSystem # timeout=10
 > git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision 67abf8278758a9ec932fab3d6393d33bf988488b (refs/remotes/origin/master)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe checkout -b master 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
Commit message: "Added gitignore file"
First time build. Skipping changelog.
[Pipeline] echo
Task 3: Displaying PROJECT_NAME Parameter...
[Pipeline] echo
The Project Name is: null
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Version 2: Numeric Logging)
[Pipeline] echo
```

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```
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters\.git # timeout=10
Fetching changes from the remote Git repository
 > git.exe config remote.origin.url https://github.com/jaswanthhitman45/LibraryManagementSystem # timeout=10
Fetching upstream changes from https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/jaswanthhitman45/LibraryManagementSystem +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision 67abf8278758a9ec932fab3d6393d33bf988488b (refs/remotes/origin/master)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe branch -D master # timeout=10
 > git.exe checkout -b master 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
Commit message: "Added gitignore file"
[Pipeline] echo
Task 3: Saving BUILD_ID to buildinfo.txt
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>echo 101  1>buildinfo.txt
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>type buildinfo.txt
101
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Version 3: Combined Parameters)
[Pipeline] echo
Task 1: Checkout latest version...
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters\.git # timeout=10
Fetching changes from the remote Git repository
 > git.exe config remote.origin.url https://github.com/jaswanthhitman45/LibraryManagementSystem # timeout=10
Fetching upstream changes from https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
```

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```
Checking out Revision 67abf8278758a9ec932fab3d6393d33bf988488b (refs/remotes/origin/master)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe branch -D master # timeout=10
 > git.exe checkout -b master 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
Commit message: "Added gitignore file"
[Pipeline] echo
Task 2: Displaying all parameters...
[Pipeline] echo
Full Details: Project=null, ID=101, App=null, Ver=1.0.0
[Pipeline] echo
Task 3: Storing in parameters.txt
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>echo PROJECT=null  1>parameters.txt

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>echo BUILD_ID=101  1>>parameters.txt

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>echo APP_NAME=null  1>>parameters.txt

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>echo VERSION=1.0.0  1>>parameters.txt
[Pipeline] echo
Final File Contents:
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>type parameters.txt
PROJECT=null
BUILD_ID=101
APP_NAME=null
VERSION=1.0.0
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

```
[Pipeline] { (Version 2: Numeric Logging)
[Pipeline] echo
Task 1: Checkout updated repo...
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters\.git # timeout=10
Fetching changes from the remote Git repository
 > git.exe config remote.origin.url https://github.com/jaswanthhitman45/LibraryManagementSystem # timeout=10
Fetching upstream changes from https://github.com/jaswanthhitman45/LibraryManagementSystem
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/jaswanthhitman45/LibraryManagementSystem +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision 67abf8278758a9ec932fab3d6393d33bf988488b (refs/remotes/origin/master)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe branch -D master # timeout=10
 > git.exe checkout -b master 67abf8278758a9ec932fab3d6393d33bf988488b # timeout=10
Commit message: "Added gitignore file"
[Pipeline] echo
Task 3: Saving BUILD_ID to buildinfo.txt
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>echo 101  1>buildinfo.txt
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters>type buildinfo.txt
101
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Version 3: Combined Parameters)
[Pipeline] echo
Task 1: Checkout latest version...
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\Library_Exp8_Parameters\.git # timeout=10
Fetching changes from the remote Git repository
```

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localhost:8085/job/LibraryManagement_Exp7_Unified/2/ws/

Jenkins / LibraryManagement_Exp7... / #2 / Workspaces

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#2'
- Timings
- Git Build Data
- Rebuild
- Open Blue Ocean
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

Workspaces for LibraryManagement_Exp7_Unified #2

- `C:\ProgramData\Jenkins\.jenkins\workspace\LibraryManagement_Exp7_Unified` on built-in